REGULATORY WORKING GROUP

Extending LoRaWAN IoT Connectivity to Satellite Networks

17th of July 2025

The LoRaWAN® standard is the leading global IoT standard for secure, carrier-grade IoT LPWAN (Low Power Wide Area Network) connectivity. This open networking specification is designed to wirelessly connect battery-operated 'things' to the internet in regional, national, or global networks. It includes key IoT requirements such as bi-directional communication, end-to-end security, mobility, and localization services.

Operating in unlicensed radio frequency bands, LoRaWAN technology is known for its exceptional long-range capabilities (up to 15 km in rural areas), low power consumption, and the ability to securely connect a large number of devices to a single network through public, private, hybrid and satellite connections enabling a scalable and flexible network for diverse IoT applications.

LoRaWAN use in unlicensed bands is indeed not limited to terrestrial connectivity. The existing ecosystem can be extended via satellites, as will be described in this document.

Extending LoRaWAN IoT Connectivity to Satellite Networks

The IoT industry is transforming connectivity, yet rural and remote areas remain underserved. Millions of IoT devices currently operate in urban regions, but the terrestrial telecommunication infrastructure leaves significant gaps in rural and hard-to-reach areas. Satellite-IoT solutions play a pivotal role in bridging this divide by providing cost effective and seamless coverage, enabling vital services in agriculture, environmental monitoring, utilities and other remote infrastructure monitoring, and marine. Moreover, they offer robust and resilient connectivity during disasters, supporting backup communications and facilitating use-cases such as early warning systems and assistance for disaster relief agencies.

The success of LoRaWAN technology over terrestrial networks has now been expanded to satellite networks, and this expansion is possible in the same frequency bands in which terrestrial networks are deployed. Lacuna Space and Plan-S have already begun using LoRaWAN technology for addressing this connectivity challenge with their satellite networks. This enables connectivity for IoT devices anywhere to the Internet through satellite networks, unlocking new possibilities for IoT applications, IoT solution providers, and terrestrial LoRaWAN operators.

Key Benefits of LoRaWAN Satellite Networks

Extending the benefits of LoRaWAN technology in unlicensed bands to satellite networks unlocks new possibilities for IoT applications, particularly in remote and underserved regions. The key advantages of satellite IoT connectivity with LoRaWAN technology are outlined below:

1. **Global Connectivity:** LoRaWAN satellite networks provide seamless coverage, especially in remote and rural areas where terrestrial infrastructure is unavailable or insufficient. This ensures that IoT devices can operate effectively anywhere on the planet.
2. **Cost-Effectiveness:** By utilizing license-exempt frequency bands and low-cost devices, LoRaWAN significantly reduces deployment and operational expenses. This makes it an ideal solution for widespread IoT adoption, even in cost-sensitive applications.
3. **Energy Efficiency:** LoRaWAN technology's low power consumption ensures device battery life, ensuring sustainability for long-term IoT operations. This is particularly valuable in inaccessible areas where battery replacement is challenging and costly.

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4. **Scalability:** Satellite-enabled LoRaWAN networks can connect millions of IoT devices, supporting diverse applications such as smart agriculture, environmental monitoring, and logistics. This scalability meets the growing demand for IoT connectivity globally.
5. **Resilient Communications:** LoRaWAN satellite networks offer robust connectivity during disasters, enabling backup communication systems. It supports essential use cases like early warning systems and disaster relief operations.
6. **Interoperability:** The technology ensures seamless integration with terrestrial LoRaWAN networks, creating a unified IoT ecosystem. This interoperability expands the range of applications and simplifies network management.

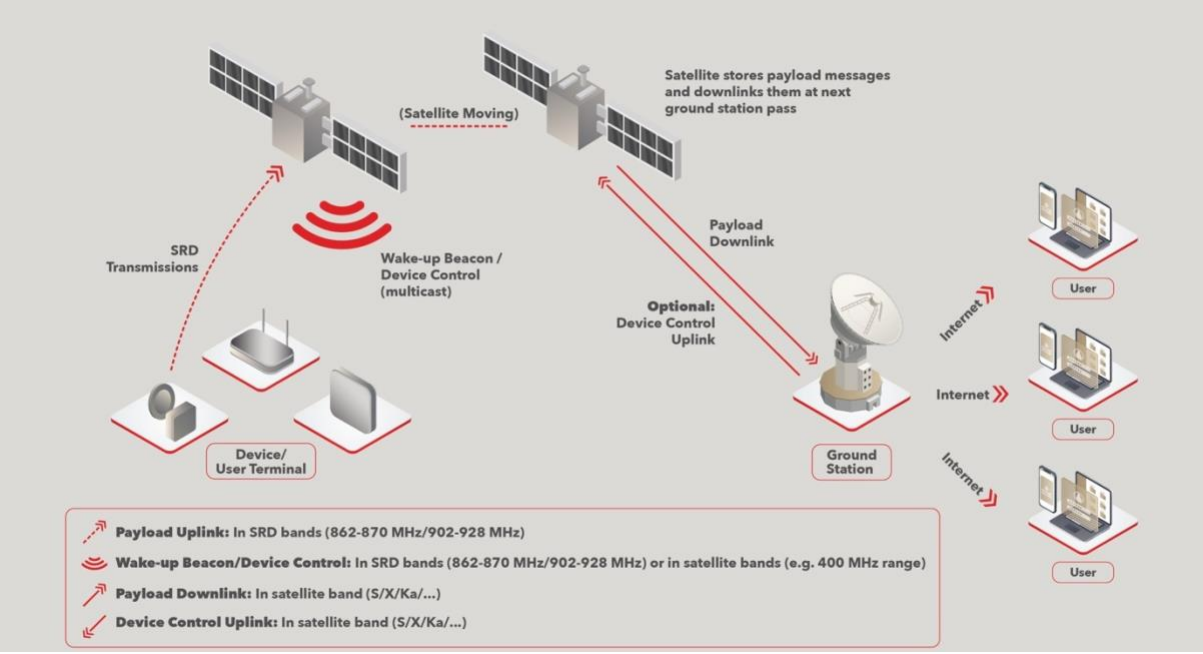
Operational Concept

The satellites receive the individual SRD transmissions from individual devices in the corresponding license-exempt bands e.g. 862-870 MHz or 902-928 MHz (depending on regional parameters and regulations) as illustrated below. The satellite then stores the received messages and – when a ground station (not necessarily in the jurisdictions where IoT devices are located) is in view – the messages are downlinked using a frequency band with a satellite service allocation.

There are not individual messages from satellites to devices in the license-exempt bands. Instead, satellites multicast signals to devices, either as wake-up beacons and/or control information to manage time synchronisation in low power modes. This assures that the impact on spectrum is minimised.

IoT devices use LR-FHSS or LoRa modulation for uplink communication, while they are configured to receive LoRa signals for wake-up or device control messages.

Satellite-SRD Operational Concept



Regulatory Framework

Satellite-IoT operations leveraging unlicensed or license-exempt bands have been validated through extensive technical studies and field trials. Significant regulatory advancements have recently been made in Europe, as

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highlighted in ECC Report 357[1], "Regulatory analyses of satellite use in the band 862-870 MHz to communicate with terrestrial SRD":

- Operation on a non-protection, non-interference basis.
- Establishment of a Power Flux Density (PFD) threshold of -142 dB(W/(m²·4kHz)) to ensure that no harmful interference is created for terrestrial users of the band.
- Extending existing terrestrial SRD frameworks to space, enabling seamless integration.

CEPT (European Conference of Postal and Telecommunications Administrations) is at the forefront of this regulatory innovation. ECC Decision (25)02, "Low power devices communicating with satellites (LPD-S) within the frequency range 862-870 MHz"[2], to harmonize and regulate the use of the 862-870 MHz band for satellite IoT across CEPT member countries has come into force after its approval at the 67th Electronic Communication Committee (ECC) meeting in Bled in June 2025.

This Decision also includes measures for the protection of the adjacent services and applications, including MFCN, with a maximum PFD limit of -146 dB(W/(m²·1MHz)) and the binding requirement for the application of ITU RR Article 17 and 18.4 for the protection of privacy of radiocommunication. This decision marks a significant step toward harmonisation of satellite use in the band 862-870 MHz to communicate with terrestrial SRD within CEPT.

The Connecta IoT Network (operated by Plan-S) and Lacuna Space are listed as compliant satellite systems under this innovative decision.

Germany's regulatory authority, BNetzA, has already implemented this model successfully. Other European administrations are expected to follow similar steps shortly, enabling this innovative use in their countries for the benefit of both citizens and industry.

These advancements mark a transformative shift in the global regulatory landscape, enabling the seamless integration of satellite IoT operations into unlicensed spectrum frameworks. This facilitates cooperation and interoperability with terrestrial LoRaWAN networks, leverages the economies of scale offered by the LoRaWAN standard, and fosters the growth of IoT ecosystems.

Showcases of LoRaWAN Satellite Networks

The techniques outlined in this paper are already implemented within operational constellations by Lacuna Space and Plan-S. These global constellations switch between the 862-870 and 902-928 MHz frequency bands as they pass over continental geo-fences to adapt to national & regional regulations in order to implement the multicast satellite-to-device operational concept as described above to achieve ultra-low power modes in remote deployments.

Below are some examples of target industrial IoT use-cases which highlight the benefits of low-power, Direct-to-Satellite (D2S) service in remote locations with limited or no terrestrial connectivity.

[1] More details can be found in the ECC Report 357

[2] The Decision can be found on the link.

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Precision Agriculture



Image Credit: Plan-S (left), Lacuna Space (right)

Utilities, including Smart Meter Reading



Image Credit: Plan-S

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Remote Infrastructure Monitoring




Image Credit: Plan-S

Marine / Maritime





Image Credit: Lacuna Space (left), Plan-S (right)

Environmental Monitoring




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Image Credit: Lacuna Space

Animal Tracking





Image Credit: Plan-S (left), Smart Parks (right)

Standardization within LoRa Alliance

The LoRa Alliance has been actively working on standardizing LoRaWAN satellite networks, addressing different types of satellite architectures and frequency bands. This standardization effort is expected to be completed within 2025, further enhancing interoperability and adoption.

LoRaWAN technology offers unparalleled advantages over its competitors, including the use of license-exempt bands, cost-effective and secure communication, long-range coverage, and exceptional energy efficiency. These benefits have now been extended to satellite networks, unlocking transformative possibilities for IoT applications, particularly in bridging the digital divide across the globe and within individual countries.

Satellite-IoT solutions effectively address this divide, providing seamless coverage that enables critical services such as agriculture, environmental monitoring, smart metering, and more. They also offer resilient connectivity during disasters, supporting backup communications and facilitating essential use cases like early warning systems and disaster relief operations.

The harmonization and regulation of unlicensed bands for satellite networks would further amplify these possibilities by enabling innovative use cases and ensuring seamless interoperability between terrestrial and satellite networks. With operators like Lacuna Space, and Plan-S already employing LoRaWAN technology in their satellite networks, IoT devices can now achieve global internet connectivity.

LoRaWAN technology's scalability, flexibility, and proven capabilities position it as a cornerstone for global IoT connectivity, ready to overcome today's challenges and power the innovations of tomorrow.

The LoRa Alliance would be very pleased to work with national regulatory authorities as well as regional telecommunication bodies in order to facilitate the adoption and the implementation of similar regulatory measures as the one adopted in Europe with the ECC Decision (25)02.

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